

Retrieval-based forecast adaptation

Executive summary of results

Adaptation pipeline and notations

For query lookback X and target $T \in \mathbb{R}^H$, the frozen backbone F produces the vanilla forecast $V = F(X)$. Retrieval returns K historical pairs (X_k, Y_k) with distance weights w_k ; their backbone forecasts are $N_k = F(X_k)$, their weighted future is $\bar{Y} = \sum_{k=1}^K w_k Y_k$, and the retrieved pairs also produce the covariate-conditioned forecast $C = F(X; \{(X_k, Y_k)\}_{k=1}^K)$. The adaptor maps $(V, C, Y_{1:K}, N_{1:K}, \bar{Y})$ to the final forecast $\hat{T}$. For a candidate adaptation A , its realized advantage over vanilla is

$$\Delta(A) = \frac{1}{H} \sum_{h=1}^H [(T_h - V_h)^2 - (T_h - A_h)^2].$$

We propose various *baselines* corresponding to different formulas fitted on $C = F(X; \{(X_k, Y_k)\}_{k=1}^K)$ to predict T , as well as *gates* which take binary decisions to choose between V and a given non-trainable adaptor $A \in \{C, \bar{Y}\}$.

Practical details

Component	Protocol
Datasets	Electricity, Traffic, Solar, and <code>exchange_rate</code> .
Settings	$(L, H) \in \{(168, 24), (336, 48), (504, 168)\}$: 12 dataset–setting configurations.
Backbone	Frozen <code>chronos2</code> .
Splits	T_0 datastore (30%); T_1 training (40%); T_2 validation (20%), T_3 eval (20%).

For a given configuration c and method m , relative improvement in nMSE over the vanilla run from the *same retrieval pipeline* is

$$I_{m,c} = 100 \frac{\text{nMSE}_{v,c} - \text{nMSE}_{m,c}}{\text{nMSE}_{v,c}} \%.$$

Tables report the mean of $I_{m,c}$ over the 12 held-out T_3 configurations.

Selected baseline and gate candidates

Code name	Family / design	Space	K	T3 gain
<code>full_ridge_shared</code>	ridge on $V, C, Y_{1:K}, N_{1:K}$	instance	3	+1.951%
<code>full_ridge_shared</code>	ridge on $V, C, Y_{1:K}, N_{1:K}$	raw	1	+1.789%
<code>cov_y_ridge_shared</code>	ridge on $V, C, Y_{1:K}$	instance	3	+1.782%
<code>full_ridge_shared</code>	ridge on $V, C, Y_{1:K}, N_{1:K}$	raw	3	+1.875%
<code>cov_avgy_ridge_shared</code>	ridge on $V, C, \bar{Y}$	instance	3	+1.765%
<code>catboost_cov_regressor_shared</code>	$G(C; g_\theta(z))_h$	raw	1	+0.031%
<code>catboost_avgy_regressor_shared</code>	$G(\bar{Y}; g_\theta(z))_h$	instance	1	-1.900%
<code>bayes_cov_shared</code>	$G(C; b(C))_h$	raw	1	0.000
<code>bayes_avgy_shared</code>	$G(\bar{Y}; b(\bar{Y}))_h$	instance	1	0.000

The corrected ranking carries each result’s retrieval identity to its matching vanilla reference. The five manually selected baseline pipelines improve mean held-out nMSE; `full_ridge_shared` with instance retrieval and $K = 3$ is best at +1.951%. The selected gates do not provide a comparable gain.

Completed current ablations

All comparisons below use the corrected retrieval-matched vanilla reference.

Axis	Equal-configuration result	Conclusion
Fourier retrieval	Selected ridges gain 1.76–1.95% with their original retrieval and 1.28–1.62% with FFT amplitude.	Do not promote Fourier.
Fixed T_0 store	Every paired ridge is at least as good with fixed retrieval: 1.80–2.09% versus 1.79–1.95% online.	Prefer the stricter fixed store.
Per-horizon ridge	Shared controls gain 1.76–1.95%; horizon fits gain 1.33–1.52%.	Extra horizon parameters hurt.
Convex fit	Matched convex variants range from -0.05% to $+0.88\%$, below every shared-ridge control.	Reject convex variants.
Δ -ridge fit	Matched variants gain 0.47–0.82%, again below every shared-ridge control.	Reject Δ -ridge variants.
Lookback L at $H = 24$	$L = 168$ gives the most balanced full-ridge gains (0.81–1.14%); $L = 504$ gives 0.24–0.41%. The $L = 24$ mean is distorted by a Traffic nMSE of 2.07×10^{10} .	Keep $L = 168$ as the stable choice.
Mixed-quantity panels	Weather worsens for every selected ridge; ETTm1 raw retrieval and the ETTm2 avgY gate show catastrophic outliers.	Keep these panels ablation-only.
Retrieval user scope	Across five selected ridges, all-user, other-users-only, and same-user-only retrieval gain 1.832%, 1.785%, and 1.480% over 60 paired cells per scope.	Cross-user candidates carry nearly all the gain; retain all users by default.

Baseline lookback-length ablation (equal mean over four datasets)

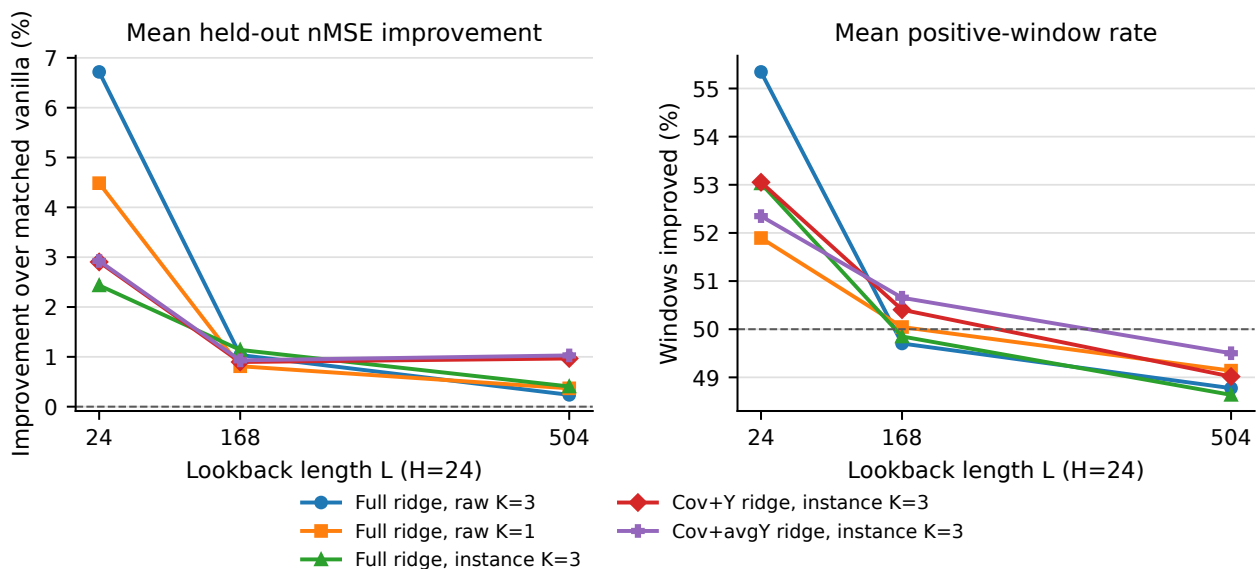


Figure 1: Completed baseline lookback ablation at fixed $H = 24$. $L = 24$ has the largest numerical means but includes the pathological Traffic scale; $L = 168$ is the balanced retained setting.

Completed follow-ups

Retrieval user scope (job 43978). The complete report contains all 324 selected adaptation manifests, and all 96 newly launched restricted-scope extractions are complete. The five ridges win 53/60, 53/60, and 51/60 dataset-setting-pipeline cells with all, other, and same-user pools. Excluding the query user costs only 0.048 percentage points of mean nMSE gain, whereas same-user-only retrieval costs 0.352 points. Raw MSE still improves by 2.335/2.275/2.148%, but MAE worsens by 0.727/0.757/0.910%, so the adaptation favours squared-error reduction rather than uniform absolute-error improvement. The gates remain weak: the best is the same-user cov CatBoost gate at $+0.219\%$, while all scoped avgY CatBoost gates are negative. Scope and candidate-pool size change together, so this supports cross-user availability rather than a causal identity effect at fixed pool cardinality.

The all-user raw- $K = 3$ control also corrects one old screen lineage. Its Electricity 504:168 predecessor consumed a smoke extraction instead of the publication profile; the valid full-profile selected-ridge mean is $+1.875\%$, not $+1.780\%$. All 18 screen rows sharing that exact extraction require a narrow rerun, but this does not affect the completed scope comparison.

TS-RAG (job 43579). All four released-model inferences completed, but the original and table-only job-43756 report stages failed to discover completed inputs across launches. The corrected completed-manifest reader rebuilt the 12-input report locally without inference. Against matching Chronos-Bolt vanilla, TS-RAG changes nMSE by $+0.44\%$ (Electricity), $+0.48\%$ (Traffic), -2.01% (Solar), and $+8.69\%$ (Exchange): $+1.90\%$ equally averaged. Against the selected full-ridge control the mean advantage is only $+0.85\%$, with losses on Traffic and Solar. The metric artifacts and rebuilt table are valid; only the cluster table stage still needs the corrected reader deployed.

Exact-split SOTA benchmark (jobs 43305, 43580, 43757). All seven selected-method rows are complete. Full shared ridge scores 0.3446/0.2500/0.3077/0.1494/0.2079/0.1060/0.0645 on ETTTh1/ETTTh2/ETTM1/ETTM2/Weather/Electricity/Exchange: mean MSE 0.2043 versus 0.2079 for locally evaluated Chronos-Bolt, 0.1970 for published TS-RAG, and 0.1910 for published Cross-RAG. It improves local vanilla on six datasets ($+1.73\%$ on the mean) and loses only on ETTm2. Job 43757's table stage exposed the same cross-launch discovery bug; the corrected seven-input report was rebuilt locally without recomputation.

K, horizon, and CatBoost sweeps (jobs 43975–43977). Instance full ridge selects $K = 5$ at $+2.00\%$; instance cov- Y and cov- $\bar{Y}$ select $K = 20$ at $+1.97\%$ and $+1.96\%$, and raw full ridge selects $K = 10$ at $+1.95\%$. For instance full ridge at $L = 504$, horizons 24/168/504 gain $+0.41/+2.78/+2.75\%$. The best CatBoost formulation is only $+0.19\%$ for the soft shared cov gate, so no learned gate is promoted.

TS-IFA results

Code name	Space / steps	T2 gain	T3 gain	Checkpoint / held-out behaviour
joint_ridge_horizon_unconstrained_full	raw / 10K	+1.84%	+0.04%	Window gains +3.26, +0.58, −3.72%; all three select 10K.
joint_ridge_horizon_unconstrained_full	raw / 30K	+3.03%	−6.71%	Selected at 29/20/20K for (168/24, 336/48, 504/168).
joint_ridge_horizon_unconstrained_full	instance / 30K	+3.25%	−17.97%	Selected at 30/26/20K for the same windows.
joint_neural_horizon_unconstrained_full	raw / instance, 30K	−	−4.60/ −2.14%	Zero T3 wins.

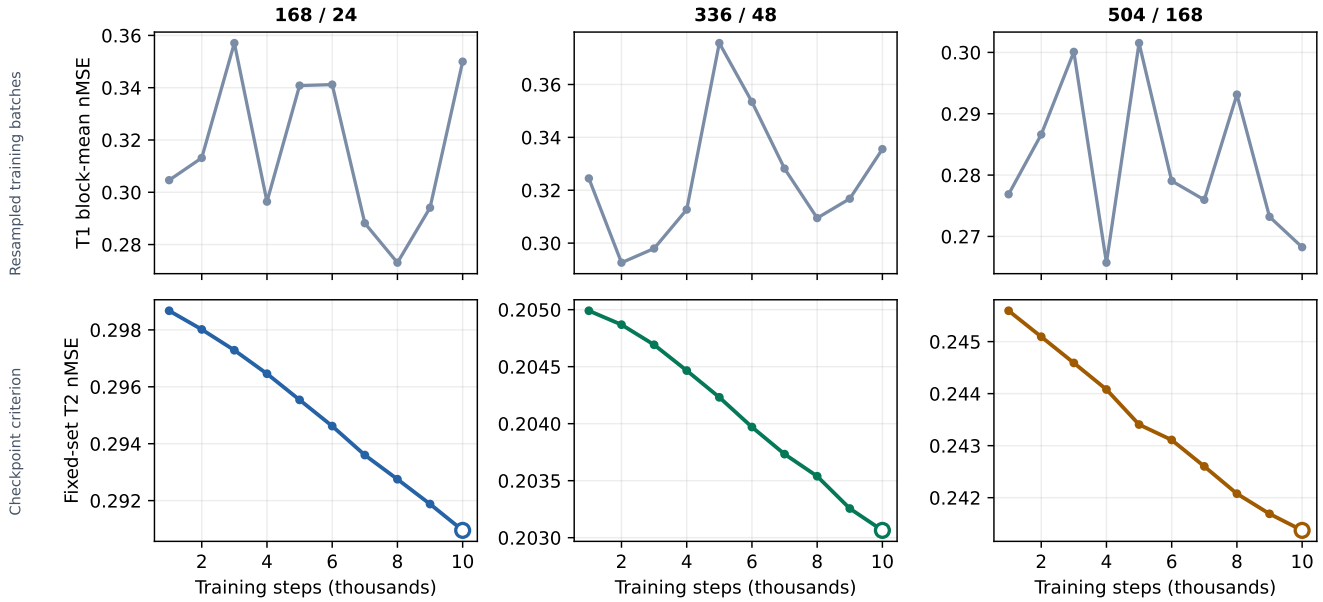


Figure 2: The 10K joint-ridge criterion curves. The gray T1 trace is a 1,000-step block mean over freshly resampled training batches, not a repeated evaluation on one fixed set, so it need not be monotone. The colored T2 trace uses a fixed set and selects 10K for all three windows.

TS-IFA is not reliable on held-out T3. Longer runs improve T2 selection but worsen T3, with the largest failures at $H = 168$; the raw 10K ridge pilot remains the least-bad aggregate result at +0.04%.